MTh301-Calculus II
Midterm Special 2006
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? At critical points function can assume
  - o maximum value.
  - o minimum value.
  - o both maximum and minimum value.
  - o **zero value**

? A plane can be perfectly determined by
  - o one point.
  - o **one point and normal vector.**
  - o one point and parallel vector.
  - o one normal vector.

? The length and width of a rectangle are measured with the errors of at most 4 % and 3%, respectively. Use differentials to approximate the maximum percentage error in the calculated area.

Solution:

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Let
x = length

y = width

A = area

Then $A = xy$ So

$$\frac{\partial A}{\partial x} = y$$

$$\frac{\partial A}{\partial y} = x$$

Therefore,

$$\Delta A \approx \frac{\partial A}{\partial x}\Delta x + \frac{\partial A}{\partial y}\Delta y$$

$$y\,\Delta x + x\,\Delta y$$

We desire percentage change in A, which is relative change multiplied by 100 so let's work out relative change first. This is given by

$$\frac{?A}{A} \approx \frac{y\,?x + x\,?y}{A}$$

$$\frac{?A}{A} = \frac{?x}{x} + \frac{?y}{y}$$

Since

$$A = xy$$

$$\frac{?x}{x} = 0.04 \quad \text{and} \quad \frac{?y}{y} = 0.04$$

The maximum possible value for $\frac{?A}{A}$ given the above constraints. This happens when $\frac{?x}{x} = 0.04$ and $\frac{?y}{y} = 0.03$, giving $\frac{?A}{A} = 0.07$. This is relative error, so the percentage error is 7%.

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Convert the Cartesian coordinates (1, 2 , 2) - into cylindrical coordinates.

Solution:

We are to convert (x,y,z) int (r, ? ,z)

Given:

$x = 1$

$y = 2$

$z = 2$

We know that:

$$r = \sqrt{x^2 + y^2}$$

Put value of x and y in the above equation:

$r = \sqrt{(1)^2 + (\sqrt{2})^2}$

$r = \sqrt{1+2}$

$r = \sqrt{3}$

$\tan? = \frac{y}{x}$

$? = \tan^{-1}(\frac{y}{x})$

$? = \tan^{-1}(\frac{\sqrt{2}}{1})$

$? = \tan^{-1}\sqrt{2}$

$? = 54.73$

$z = -2$

So the cylindrical coordinates are $(\sqrt{3}, 54.73, -2)$

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$a = +\hat{i} - \hat{j} + \hat{k}$

? A unit vector in the direction of is

- $\frac{\langle \sqrt{1}, \sqrt{1}, -\sqrt{1} \rangle}{\langle -3, -3, 3 \rangle}$
- $\langle \frac{3}{\sqrt{1}}, \frac{-3}{\sqrt{1}}, \frac{3}{-\sqrt{1}} \rangle$
- $\langle \frac{-}{2}, \frac{}{2}, \frac{}{2} \rangle$
- $\langle \frac{}{2}, \frac{}{2}, \frac{}{2} \rangle$

Find equation of the tangent plane to the surface $z x = +^3 y^3$ at the point $(2, 1, 3)$.

Solution:

$f_x = 3x y$

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$f_y = 3yx$

$f_z = -1$

$fP_x( ) = - = 6\ 1\ 5$

$fP_y( ) = - = 3\ 2\ 1$

$fP_z( ) = -1$

Equation of tangent line to the surface through P is:

$5(x-2) + (1)(y-) + (-)(z-3) = 0$

$5x + 10 - y + 3 = 0$

Hence:

$5xyz + - - = 0$

8

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Dot product of $a = +i\ 2j\ k$ and $b = +2i\ 3j + 2k$ is

- 8
- 5
- **10**
- 13

? Evaluate the double integral $\iint_R x^2\,dA$; R is the region bounded by $y = 16, y = x, x = 8$.

Solution:

Area of $R = \iint_R x\,dA$

$= \int_8^{16} \int_x^{x^2} x\,dy\,dx$

$= \int_8^{16} x^2 - x\,dx$

$= \int_8^{16} x^3$

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$= \int_{2}^{x} - 16x dx$

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$$= \int_{2}^{8} \frac{x^4}{4} - \frac{16x^2}{2}$$

$$= \int_{2}^{8} \frac{x^4}{4} - 8x^2$$

$$= \int_{2}^{8} \frac{x^4}{4} - \frac{2^4}{4} \times 8^2 - 8 \times 8^2$$

$$= \mathbf{972}$$

? Domain of $x + y$ is

- $x y + > 0$
- $x = 0 , y = 0$
- $x > 0, y > 0$
- $x y + = 0$

? Use double integrals to compute the volume of the solid bounded by the *cylinder* $x^2 + y^2 = 9$ *and the planes* $y z + = 9$ *and* $z = 0$

Solution:

$$\iint_R (9 - y)\, dA$$

R as a type 1 region we find that:

$$V = \int_{-2}^{2} \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (9 - y)\, dy\, dx$$

$$= \int_{-2}^{2} \left[9y - \frac{y^2}{2} \right]_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dx$$

$$= \int_{-2}^{2} 18\sqrt{9 - x^2}\, dx$$

$$18 \cdot \frac{1}{2} \cdot \pi \cdot 9^2 = 729\pi$$

$$= \quad ? \qquad ? \ 2$$

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